

DSP Non-Stationary Nature of Speech Signal

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Some Good Books in Speech Processing

- L. R. Rabiner, B. H. Juang and B. Yegnanarayana, "Fundamentals of speech recognition", Pearson Education, 2009.
- L.R. Rabiner and R.W. Schafer, "Digital Processing of Speech Signals", Pearson Education, Delhi, India, 2004
- J. R. Deller, Jr., J. H. L. Hansen and J. G. Proakis, "Discrete-Time Processing of Speech Signals", Wiley-IEEE Press, NY, USA, 1999.
- D. O'Shaughnessy, "Speech Communications: Human and Machine", Second Edition, University Press, 2005.
- T. F. Quatieri, "Discrete time processing of speech signals", Pearson Education, 2005.



Stationary vs Non-Stationary Signals

- Understand difference between stationary and non-stationary signal
- Get feel about the non-stationary nature of speech signal
- Fourier transform based frequency domain analysis
- Understand limitations of Fourier transform in case of non-stationary signals



Review of Signals, Systems & Processing

- Signal is a physical quantity that is measurable
- System is a physical entity that exists
- Signal is produced from a system
- Classification of signals : continuous v/s discrete, periodic v/s aperiodic, energy v/s power, periodic v/s aperiodic, deterministic v/s random, stationary v/s non-stationary ...
- In a first level signals, systems and processing related courses, emphasis was not provided on the stationary v/s non-stationary classification
- Most of the signals, systems and signal processing concepts were taught based on the implied assumption of stationarity.

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Motivation for Non-Stationarity

- Speech signal deviates in this aspect
- Speech is an example for non-stationary signal where as conventional synthetic signals like sine wave, triangular wave, square wave and so on are stationary in nature
- Hence different approaches and tools are needed to process the speech signal
- To appreciate new approaches and tools developed exclusively for speech processing, we should get a feel about stationary v/s non-stationary classification

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Stationary Signal

- A signal is said to be stationary if its frequency or spectral contents are not changing with respect to time.
- Sine wave is stationary signal as its frequency will not change with time.
- Suppose if you change the frequency, then it altogether becomes a new sine wave.
- Stationarity should not be confused with time varying amplitude in the time domain as in the case of sine wave.
- Stationarity is linked to the behavior of the frequency contents of the signal with respect to time and nothing else.

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Fourier Transform

- Frequency domain representation
- Continuous time Fourier Transform (CTFT)
- **CTFT** : $X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$
- **Inverse CTFT** : $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)e^{j\omega t} d\omega$
- Given CT aperiodic signal $x(t)$, we can do frequency domain analysis using CTFT pair of relations

CTFT of Periodic Signals

- Let $x(t) \leftrightarrow X(j\omega) = 2\pi\delta(\omega - \omega_0)$
- $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi\delta(\omega - \omega_0)e^{j\omega t}d\omega = e^{j\omega_0 t}$
- Multiplication of a function with an impulse
- Sampling/shifting property of an impulse
- Next, let $x(t) \leftrightarrow X(j\omega) = \sum_{k=-\infty}^{\infty} 2a_k\pi\delta(\omega - k\omega_0)$
- $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$
- Thus $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = X(j\omega) = \sum_{k=-\infty}^{\infty} 2a_k\pi\delta(\omega - k\omega_0)$
- LHS is CTFS coefs a_k based and RHS CTFT representation.
- Periodic signal $x(t)$ represented by CTFT as impulse train in FD, each impulse $(\omega - k\omega_0)$ weighted by respective Fourier coef a_k



CTFT of Impulse Train in TD

- Let $x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$ with period T
- $X(j\omega) = \sum_{k=-\infty}^{\infty} 2a_k \pi \delta(\omega - k\omega_0)$
- $a_k = \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-jk\omega_0 t} dt = \frac{1}{T}$
- $X(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - \frac{2\pi k}{T})$
- CTFT of a periodic impulse train in TD with period T is a periodic impulse train in FD with period $\omega_0 = \frac{2\pi}{T}$
- Inverse relationship between time and frequency domains

Discrete Time Fourier Transform

- Frequency domain representation of DT signal $x(n)$
- Discrete time Fourier Transform (DTFT)
- **DTFT** : $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}$
- **Inverse DTFT** : $x(n) = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega$
- Like CTFT, DTFT can also be used for both aperiodic and periodic signals or sequences



Deriving DFT Pair of Relations

- Frequency domain representation of DT signal $x(n)$
- Discrete time Fourier Transform (DTFT)
- **DTFT** : $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}$
- **Inverse DTFT** : $x(n) = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega$
- Let $x(n)$ is a DT sequence block of N samples
- For time domain alias free condition, $N_s \geq N$ and let $N_s = N$
- Any frequency range of 2π say $-\pi$ to π is sampled into $N_s = N$ samples as $1/N_s$ will be interval between successive samples.
- Let $\omega = \frac{2\pi k}{N}$, where $k = 0, 1, \dots, (N-1)$

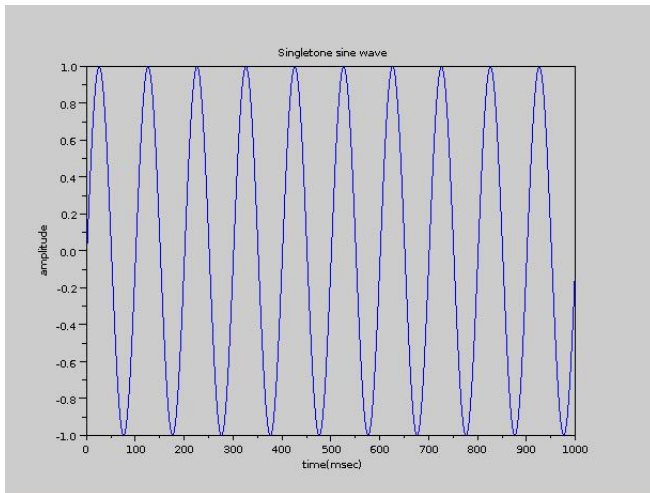


Deriving DFT Pair of Relations

- Any frequency range of 2π say $-\pi$ to π is sampled into $N_s = N$ samples as $1/N_s$ will be interval between successive samples.
- Let $\omega = \frac{2\pi k}{N}$, where $k = 0, 1, \dots, (N - 1)$
- When sampled like this, ω becomes discrete.
- Discrete Fourier Transform (DFT)** : $X(k) = \sum_{n=0}^{N-1} x(n)e^{-j\frac{2\pi kn}{N}}$, where $k = 0, 1, \dots, (N - 1)$
- Accordingly, 2π maps to N and integration in inverse DTFT to summation over N interval.
- Inverse Discrete Fourier Transform (DFT)** : $x(n) = \sum_{k=0}^{N-1} X(k)e^{j\frac{2\pi kn}{N}}$, where, $n = 0, 1, \dots, (N - 1)$

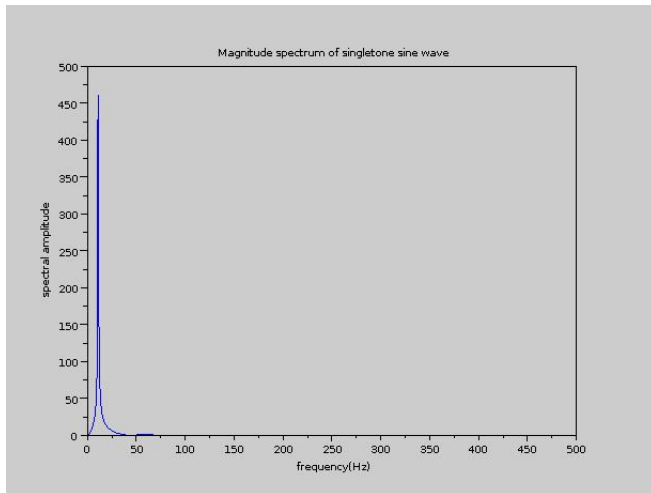


Single tone Sine wave in Time Domain



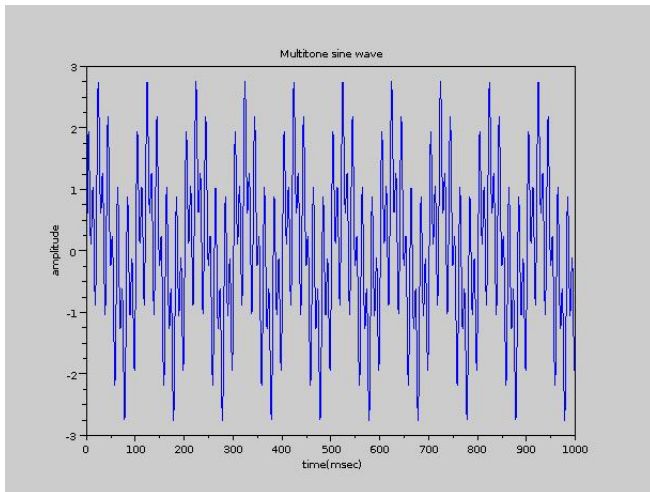
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Single tone Sine wave in Frequency Domain



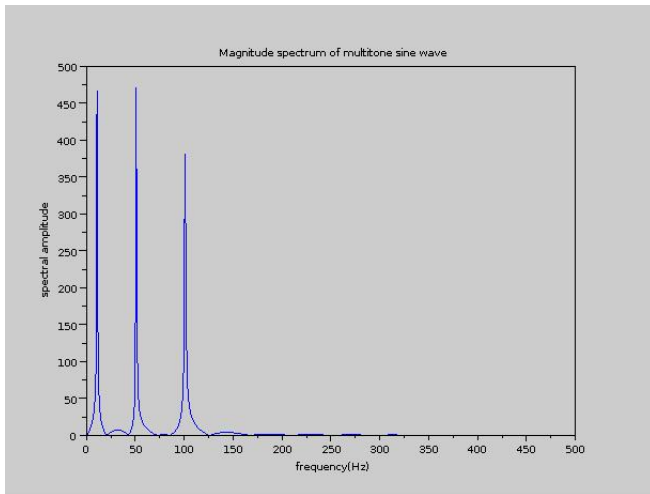
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Multitone Sine wave in Time Domain



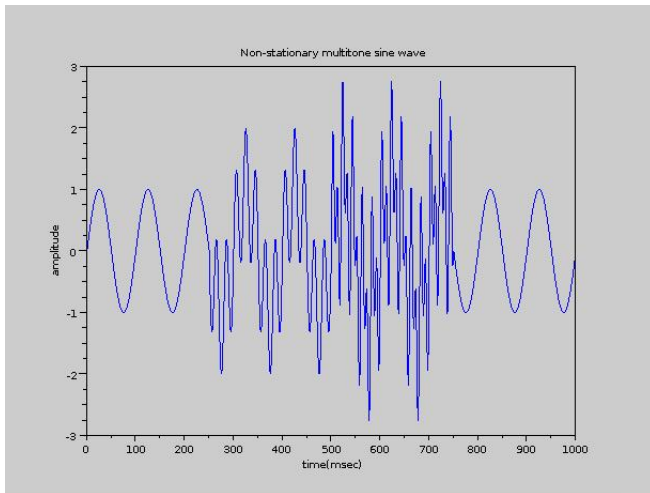
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Multitone Sine wave in Frequency Domain



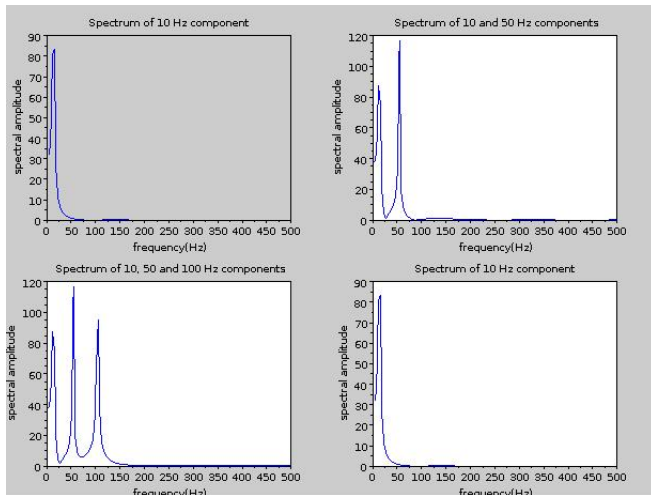
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Multitone Non-Stationary Sine wave in Time Domain



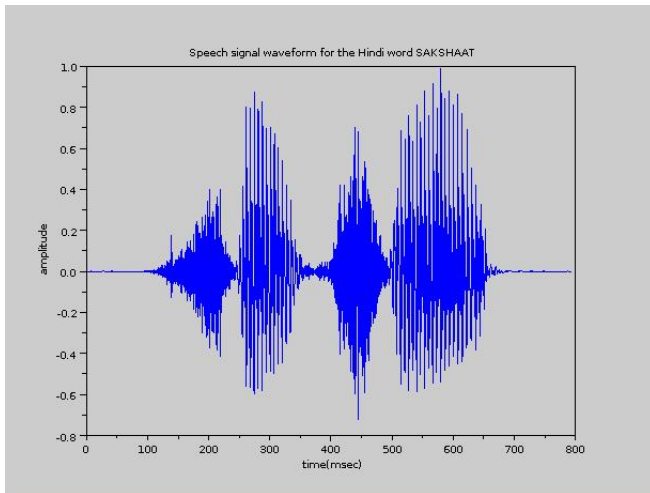
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Multitone Non-Stationary Sine wave in Frequency Domain



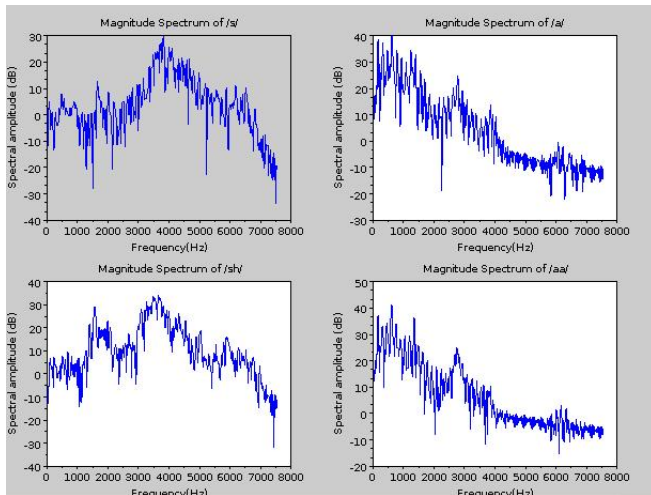
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Non-Stationary Speech in Time Domain



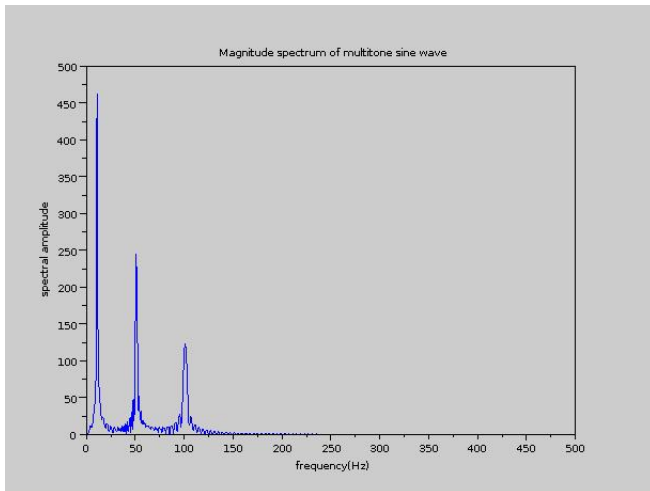
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Non-Stationary Speech in Frequency Domain



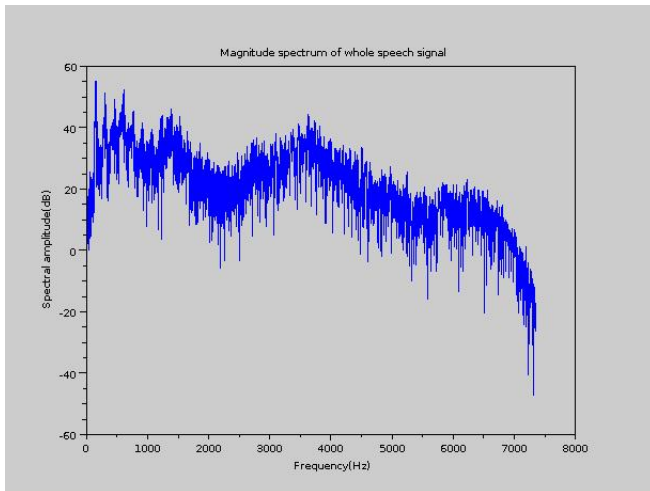
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Limitation of Fourier Representation in Non-Stationary Multitone Sinewave



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Limitation of Fourier Representation in Non-Stationary Speech



Taken from public domain. Original authors are highly acknowledged.